Multimedia Signal Processing - 2nd Module

September 7, 2023

1. **Question (7 pts)**
   Consider a continuous source X that emits symbols following a probability distribution function $p_X(x)$.
   (a) Consider the following series of symbols emitted by X

   $$x = \{0.1,\ -0.9,\ 0.7,\ -0.7,\ -0.7,\ 0.2,\ -0.7,\ -0.3,\ 0.8,\ -0.2,\ 1.5\}.$$

   Design a Lloyd-Max quantizer tailored to these symbols using $M = 3$ decision regions and the thresholds $\{-\infty, -0.5, 0.5, +\infty\}$ as initialization. Stop after one full iteration.
   (b) Apply the obtained quantizer to x thus obtaining $\tilde{x}$.
   (c) Compute the entropy of the source representing $\tilde{x}$ estimating the alphabet and the probability mass function from the data you have.
   (d) Derive an Huffman code to transmit $\tilde{x}$.
   (e) Considering the obtained Huffman code, compute the average code length explaining its meaning.
   (f) What happens if you still use the same Huffman code after you notice that the probability mass function you have estimated is wrong?

2. **Question (4 pts)** Consider the problem of image coding.
   (a) Report the block diagram of the JPEG encoder.
   (b) Report the block diagram of the JPEG decoder.
   (c) Describe all the blocks of the diagrams in order to explain how JPEG coding scheme works.

3. **Question (4 pts)** Considering the process of defining a codebook for the symbols emitted by a DMS source.
   (a) Define what do we mean by "code".
   (b) Provide the definition of one-to-one, instantaneous, and uniquely decodable codes.
   (c) Consider a source with an alphabet with cardinality equal to 3. Provide an example of code that is uniquely decodable.

4. **Questions (8 pts - each answer can be either TRUE or FALSE - Incorrect answers give you -1 points, unanswered 0 points, correct answers 1 point - Provide a brief comment when in doubt)**

   - [T] [F] Consider a perceptual audio coding scheme. If a pure tone is characterized by a Signal to Mask Ratio of 12 dB, we can expect to roughly allocate 2 bits for its coding to avoid introducing quantization noise.
   - [T] [F] It is possible to encode a DMS source X with more distortion than σ_X^2 (i.e., the variance of the source).
   - [T] [F] A closed-loop DPCM is always preferable to a normal PCM scheme.
   - [T] [F] Given a video coding scheme, the decoder does not need to know which motion estimation algorithm has been used by the encoder.
   - [T] [F] Given a discrete source, the average code length of the Shannon code can be smaller than the average code length of the Huffman code.
   - [T] [F] If lossy coding is applied to a source (e.g., quantization), it is useless to apply a further step of lossless coding.
   - [T] [F] Mutual information between two discrete sources X and Y cannot be negative.
   - [T] [F] Consider a vocoder exploiting LPC. It is not safe to quantize LPC coefficient directly, as it can cause instability.

5. **Question (Matlab - 10pts)**

   Consider you have an image file called `peppers.png` in your Matlab working directory. This is a RGB image with resolution 512×384.

   Data loading

   (a) Load the image and display it.

   (b) Considering that this is an RGB image stored as 8 bit unsigned-integers (uint8), compute how much space is needed to store the uncompressed image in memory.

   (c) Extract the red color component from the image.

   DPCM coding Considering the color component you extracted from the image, implement a close-loop DPCM-based coding scheme. Consider the image as a monodimensional vector, use the previous pixel as predictor, and assume that the first pixel is known by the decoder.

   (d) Turn the two dimensional image into a monodimensional vector.

   (e) Consider that you want a quantizer working with 16 levels, define a quantizer that is reasonably adapted to the residual dynamic (hint: use an open-loop estimate of the residual to tune your quantizer).

   (f) Encode and decode the vector using the defined DPCM and quantizer.

   (g) Turn the decoded vector back into a matrix and display the decoded image.

   (h) Compute the introduced MSE.

   (i) Compute the entropy of the encoded data.